Customer Retention & Repeat Purchase Analysis

E-commerce beauty store – sample dataset, 2020–2023

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github.com/annguyen68-crypto/customer-retention-repeat-purchase-analysis

Executive summary

An online beauty store attracts many first-time buyers, but most never come back. This report measures how many customers return and when, finds where they drop off on the way to a repeat purchase, groups customers by buying behaviour, and designs an A/B test to check whether a second-order voucher would pay off.

- **69% of 17,415 customers placed only one order.** In most quarterly cohorts, only about 4–7% of new customers buy again in the following quarter.
- **The biggest loss is between the 1st and 2nd order** (almost 12,000 customers). After that, each step keeps about one third of customers.
- **Customers come back slowly:** only 4.6% place a 2nd order within 90 days; the median time to the 2nd order is 402 days.
- **Revenue is concentrated:** Champions and Loyal customers are 11.3% of customers but 27.2% of revenue.
- **Repeat buyers are more profitable** (profit per order about 56.8 vs 38.8) and receive smaller discounts (12% vs 15%).
- **Recommendation:** focus on moving customers to a 2nd order, and test a voucher before rolling it out. A 15 USD voucher is the only option that can be tested within a year with the current number of new customers.

Data and approach

Item	Details
Data	Sample e-commerce dataset: 51,290 sales lines, 25,739 orders, 17,415 customers, 01/2020 – 12/2023; 4 tables (EcomSales, Customer, Product, Region)
Cleaning	Checked missing values, duplicates and outliers; standardised join keys with repeated prefixes (e.g. PPP000002 → P000002) so every sales line matched its reference table; converted dates
Methods	Quarterly cohort retention, repeat-purchase funnel, time to 2nd order (right-censoring handled), RFM segmentation, profile of repeat buyers, A/B test design (break-even MDE, power analysis)
Tools	Python (Pandas, NumPy, Matplotlib, Seaborn, statsmodels), Jupyter, VS Code

1. Most customers buy only once

Out of 17,415 customers, 11,979 (69%) placed a single order and 89% placed one or two. Only 201 customers (1.2%) placed five or more orders. Keeping customers after their first purchase is the store's main problem.

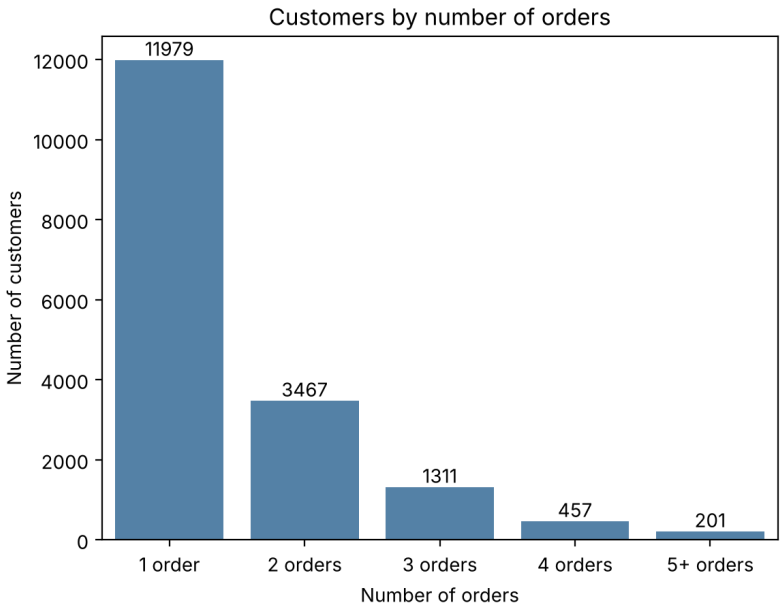


Figure 1. Customers by number of orders

2. Cohort retention

Customers are grouped by the quarter of their first purchase. Retention one quarter later is only about 4–7% across cohorts and stays around 3–10% in later quarters without a steady decline. Q1 cohorts are the smallest and Q4 cohorts the largest, but Q1 customers return just as often. Corporate and Consumer customers show almost the same pattern.

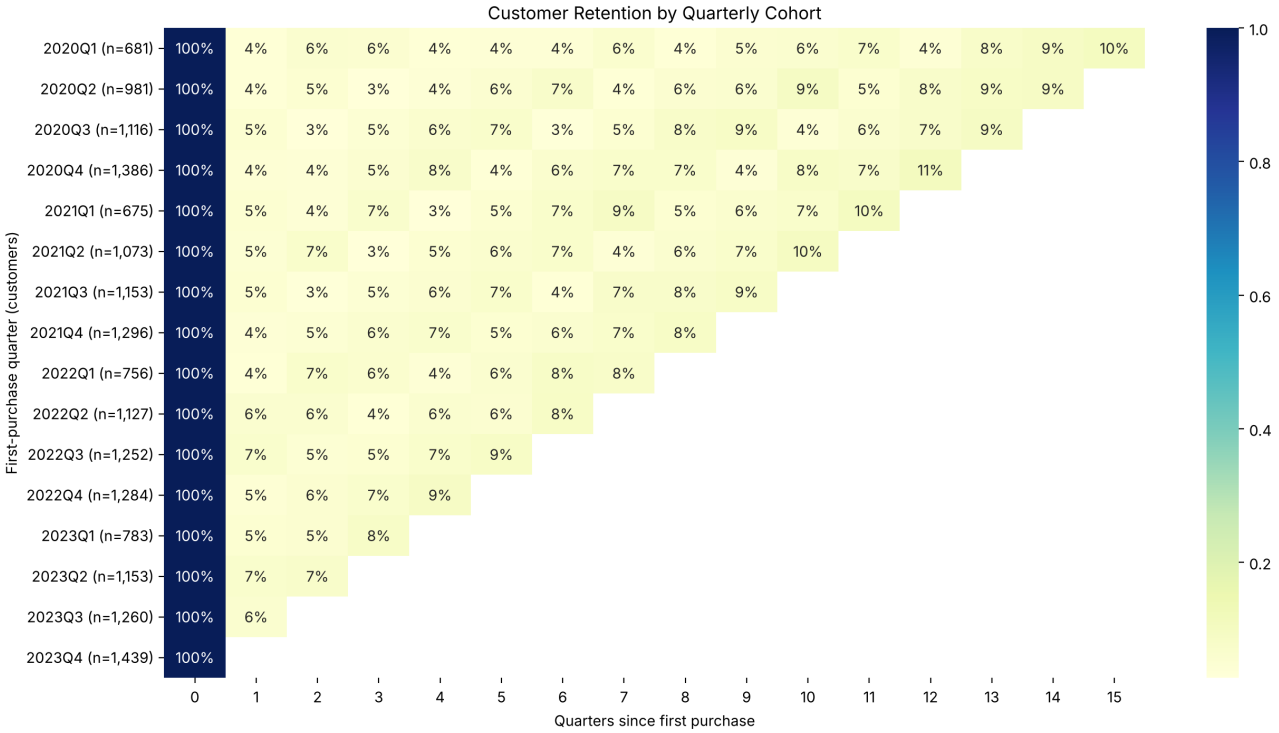


Figure 2. Share of each quarterly cohort that buys again in later quarters (cohort size in brackets; recent cohorts have fewer quarters of history)

3. Repeat-purchase funnel and time to the 2nd order

Each step of the funnel keeps roughly one in three customers (31%, 36%, 33%, 31%), so no single step is unusually weak. However, the 1st → 2nd step loses by far the most customers in absolute terms, so improving it has the largest impact.

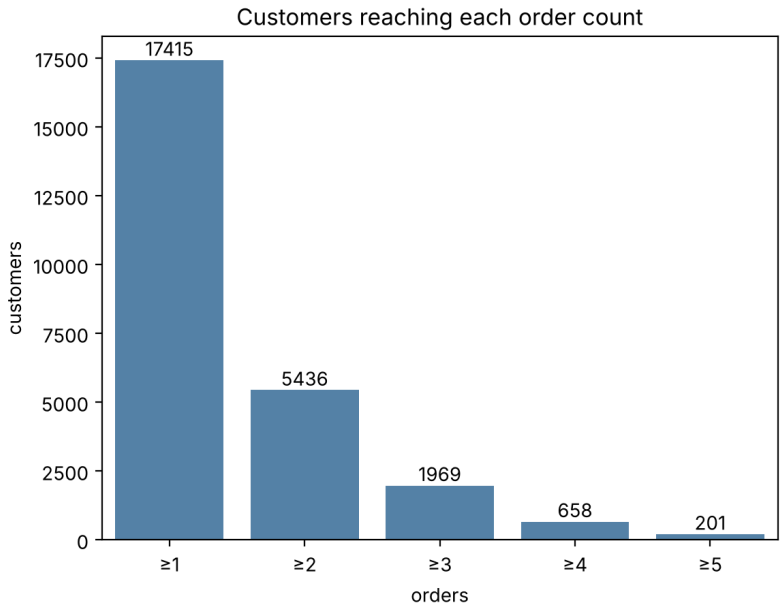


Figure 3. Customers reaching each order count

To avoid understating repurchase for customers who joined near the end of the data, timing is measured only on the 12,780 customers with at least 12 months of history:

Window after 1st order	Placed a 2nd order
Within 90 days	4.6%
Within 180 days	9.0%
Within 365 days	17.8%

Among customers who did come back, the median time to the 2nd order is 402 days (middle 50%: 195–677 days). Customers return slowly and at no fixed moment, so a single "best time" for a reminder does not exist. Customers who joined earlier had more time to return, so this median describes this dataset rather than a fixed rule.

4. RFM segmentation

Customers are scored on Recency (R) and Frequency (F); Monetary value (M) describes each segment. Because 69% of customers have only one order, segments are assigned with R and F rules instead of the standard 3-digit RFM code.

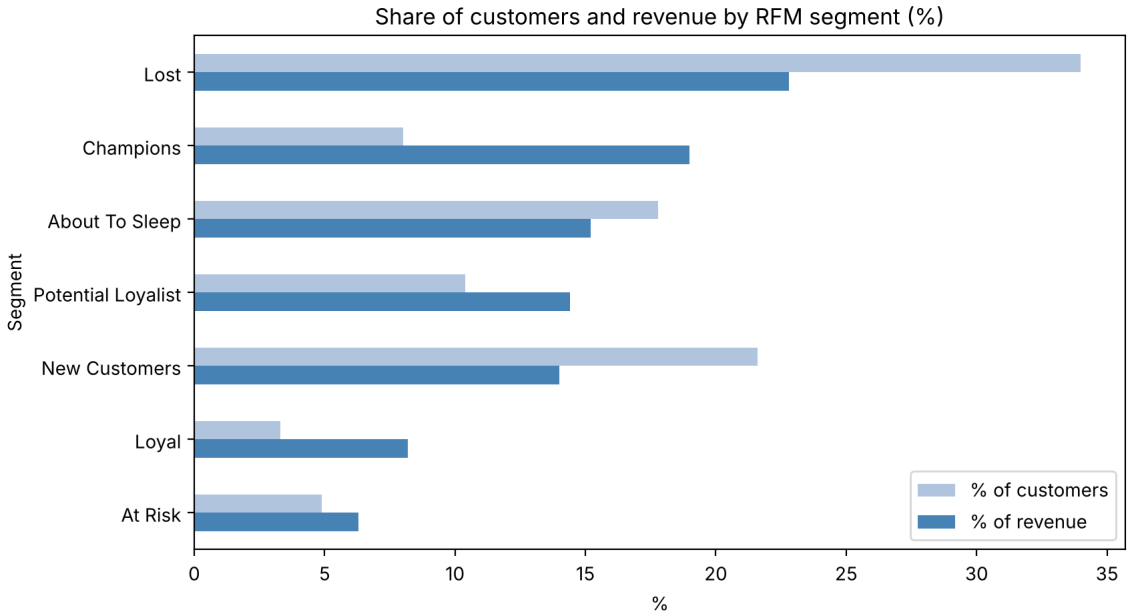


Figure 4. Share of customers and revenue by RFM segment

Segment	Customers	Revenue	What it means
Champions	8.0%	19.0%	Recent and frequent (3.5 orders, last purchase ~108 days ago): retain
Loyal	3.3%	8.2%	Frequent but inactive for ~519 days: win back
New Customers + Potential Loyalist	32.0%	28.4%	Bought recently (1–2 orders): best target for a next-order offer
Lost	34.0%	22.8%	One order, long inactive, low spend: low-cost channels only

5. What repeat buyers look like

Average profit per order rises with the number of orders (about 38.8 for one-order customers to 56.8 for 5+ orders), and profit margin rises from 15.6% (one order) to about 19% (four or more orders), while the average discount falls from 15% to 12%. Repeat buyers are not attracted by bigger discounts. This is an association, not proof of cause, which is why an experiment is needed.



Figure 5. Average profit per order by number of orders

Income does not separate repeat buyers from one-time buyers (average income 55,000–58,000 in every group), and the product mix of one-time buyers is almost identical to everyone else's (Body care is about 51% of units in both). Customers should therefore be targeted by buying behaviour, not by income or category. Face care has the highest share of customers with 3+ orders (about 24% vs 15% for Body care), but this partly reflects how many customers each category reaches, so it is a hypothesis to test rather than a conclusion.

6. A/B test design: second-order voucher

Hypothesis: new customers who receive a voucher after their first order are more likely to place a 2nd order within 90 days. **Design:** random 50/50 split into control (no voucher) and treatment (voucher); primary metric is the 90-day 2nd-order rate; secondary metric is profit after the voucher cost.

The minimum detectable effect (MDE) is set at the **break-even lift**: the smallest increase that makes the voucher profitable, $p_1 = p_0 \times P / (P - V)$, where p_0 = 4.6% (baseline), P = 41.4 (average profit per order) and V = voucher value. Sample size uses alpha = 0.05 and power = 0.8; duration assumes about 1,088 new customers per quarter plus a 90-day follow-up. The break-even rule assumes every treatment customer who places a 2nd order uses the voucher and that a 2nd order earns the average profit per order.

Voucher	Break-even rate	MDE	Relative lift	Customers per group	Duration
5 USD	5.3%	+0.6 pp	+13.7%	18,187	~34 quarters
10 USD	6.1%	+1.5 pp	+31.9%	3,641	~7.7 quarters
15 USD	7.3%	+2.6 pp	+56.8%	1,250	~3.3 quarters

A small voucher is easy to break even on but takes years to measure; a large voucher is quick to measure but needs a much bigger lift to pay off. Only the 15 USD option gives a result within a year.

7. Recommendations

- Focus on moving customers from the 1st to the 2nd order with reminders or vouchers, since this step loses the most customers.
- Test the 15 USD voucher first (about 10 months). Roll it out only if the treatment group beats the control group ($p < 0.05$) and exceeds the 7.3% break-even rate.
- Target offers by behaviour (RFM): retain Champions, win back Loyal customers, push New Customers and Potential Loyalists to their next order, and use only low-cost channels for Lost customers.
- Test Face care as an introductory product for new customers and measure the 2nd-order rate (hypothesis to be tested).

8. Limitations

- The data ends in 2023 Q4, so recent cohorts have little history; customers who first bought near the end have not had time to return.
- The analysis shows associations, not causes; the A/B test is a design only, with no experiment results.
- There is no information on channels, marketing spend or promotions, so seasonal explanations are hypotheses.
- The 5+ order group is small (201 customers), so conclusions about it should be treated with caution.